

Week 2 Progress Report and Mathsy Things

Hydroelastic Floe Identification: Early Literature Review and Pipeline Development

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Chapter 1

Week 2 progress report and mathsy things

1.1 Introduction and Regime Framework

This section discusses my main pipeline ideas for extracting hydroelastic properties from the inertial motion unit (IMU) time series measurements, along with a discussion on some of the math content I have come across. I am going to try keep the tone conversational in efforts of not making this a boring read, as it will not feature in the final report.

Where there are maths/process elements I am unsure of, I will make it obvious and hopefully our next meeting can discuss some of these issues.

The 2022 Antarctica cruise had a series of wave measurement buoys deployed in efforts of reconstructing the **directional wave spectra** of the marginal ice zone. They were equipped with 6 degree of freedom IMU's (and hence able to measure each buoy's heave, surge, sway, pitch, roll and yaw). For specifics on the actual wave buoy utilized, see [?].

From their gathered data set (along with every dataset gathered from ice tethered buoys) they planned on extracting major wave statistics such as significant wave height and mean period. An issue with this is that the response of the buoy to incident waves differs depending on what structure the buoy is on. A buoy in the open ocean will follow the waves with a relative amplitude operator (RAO) of approximate unity (explained in next section), while a buoy on an ice sheet will perceive a damped response, along with vibration induced higher order modes from ice flexure.

Thus it is essential to know properties of the geometry the buoy is on to infer accurate wave statistics. The goal of this project is to solve the inverse problem: given the buoy's response to waves, can hydroelastic properties of the buoy-ice regime be inferred?

1.1.1 The 3 major regimes

This section proposes a generic 3 - class regime (where the major dynamics change):

1. **Open Ocean:** Rigid body dynamics dominates
2. **Intermediate floe:** Flexural response dominates
3. **Large, consolidated sheet:** High pass behaviour, flexure, extreme damping (through high diffraction/scattering)

My main interest is in the 2nd item; intermediate floes. This is where the major hydroelastic properties of the floe dominate the response, and finding these from the inertial signature is most compelling. The rigid body response is very "easily" identifiable, as the geometry of the buoy is known thus the system becomes deterministic for a given incident spectrum.

Another reason the intermediate floe response interests me is because the buoy has a single IMU, essentially tracking one particle on the surface of the floe. Thus it is unlikely to be possible to infer a full geometry of the floe, but there should be an identifiable set of geometries or more interestingly, hydroelastic properties which can match the response.

The next section explains some theory governing each regime (that I have researched, whether in this project or my previous project where I designed a wave buoy).

1.2 Rigid body dynamics

Aside on notation: I like to easily differentiate between scalars (no underline), vectors (single underline OR bold) and matrices (double underline). If the notation is inconsistent or confusing I can clarify.

I am going to assume (probably incorrectly) that you know very little about rigid body dynamics with respect to floating bodies. I will attempt to explain the theory, which in no means is very complicated, as best I can. I direct you to [?] if you would like a more in depth read (really nice book on marine hydrodynamics). I do understand this section very well and have extensively dealt with it, so I will only provide a summary of the main concepts for now.

Let the motion of a buoy be captured by its 6 DOF generalized position vector $\xi \in \mathbf{R}^6$. Applying Newton's 2nd law allows for a fully deterministic set of equation for the motion of a buoy external wave forcing:

$$(\underline{\underline{M}} + \underline{\underline{A}})\ddot{\xi} + \underline{\underline{B}}\dot{\xi} + \underline{\underline{C}}\xi = \mathbf{F}_{\text{exc}} \quad (1.1)$$

Here each matrix is $\in \mathbf{R}^{6 \times 6}$.

1. $\underline{\underline{M}}$ is the rigid body matrix. This can be thought of as "mass". For a symmetric body with the origin chosen at its centre, the main diagonal holds the most prevalent information, being a function of mass, metacentre heights, displaced volume and moments of inertia. For a buoy of known geometry and material properties, this is trivial to solve for.
2. $\underline{\underline{A}}$ is the added mass matrix. Following potential flow theory, one can think of the added mass as the mass of water a buoy must displace as it moves (in our case we care most about heave). It is a **frequency** dependent matrix, and most commonly solved for using a boundary element method solver. Inputs often include a mesh of the geometry along with spectral forcing.
3. $\underline{\underline{B}}$ is the potential damping matrix. It effectively acts as a damper to the system by taking into account the energy the buoy loses by creating radiating waves outwards as it moves. Also **frequency** dependent.
4. $\underline{\underline{C}}$ is the hydrostatic restoring matrix. Effectively the mechanical spring matrix.

Note that the above equation is a One thing this model does not account for is a phenomenon known as the **fluid memory effect**. This describes how the hydrodynamic radiation force acting on a floating body depends on the entire history of its motion, as waves radiated by previous

motions linger on the free surface and continue to exert forces on the body as they decay. This effect was captured by W.E. Cummins [?], and further made numerically stable and efficient for MATLAB simulations by Thor Fossen and Tristan Perez [?]. These fluid memory effects are incorporated into the governing equation to yield the Cummins equation:

$$(\underline{\underline{M}} + \underline{\underline{A}}_\infty)\ddot{\xi}(t) + \int_0^t \underline{\underline{K}}(t - \tau)\dot{\xi}(\tau) d\tau + \underline{\underline{C}}\xi(t) = \mathbf{F}_{\text{exc}}(t) \quad (1.2)$$

where $\underline{\underline{A}}_\infty \in \mathbf{R}^{6 \times 6}$ represents the infinite-frequency added mass matrix, defined as the instantaneous fluid inertia felt during sudden acceleration before surface waves can form:

$$\underline{\underline{A}}_\infty = \lim_{\omega \rightarrow \infty} \underline{\underline{A}}(\omega) \quad (1.3)$$

The term $\underline{\underline{K}}(t) \in \mathbf{R}^{6 \times 6}$ is the **fluid memory kernel** (also known as the retardation function matrix or impulse response function). It accounts for the time-decaying forces exerted on the body by surface waves radiated from its past motion. It is evaluated directly from the frequency-domain potential damping matrix $\underline{\underline{B}}(\omega)$ via the inverse Fourier cosine transform:

$$\underline{\underline{K}}(t) = \frac{2}{\pi} \int_0^\infty \underline{\underline{B}}(\omega) \cos(\omega t) d\omega \quad (1.4)$$

By Ogilvie's relations [?], the frequency-dependent added mass $\underline{\underline{A}}(\omega)$ can similarly be related to the memory kernel via:

$$\underline{\underline{A}}(\omega) = \underline{\underline{A}}_\infty - \frac{1}{\omega} \int_0^\infty \underline{\underline{K}}(t) \sin(\omega t) dt \quad (1.5)$$

Note that the Cummins equation is a technically inconsistent integro-differential equation: it involves a time-convolution of the frequency dependent memory kernel with the time varying velocity.

All of these quantities are fully discoverable for a given geometry mesh. Boundary element methods/paneling can be used to calculate all of these constant/frequency dependent quantities. For high accuracy, CFD is a possible option, or experimental measurements. A boundary element solver is also capable of outputting these matrices, but for simple geometries there exist analytic solutions. I have a lot of resources to tackle this section, so I would rather focus on the next 2 regimes. Another of Thor Fossen's contributions is a full MATLAB toolbox [?] which contains many functions allowing for the simulation of vessel (or in our case floe) motion. It pairs with his textbook, and thus is very easy to use and modify.

An important concept is the RAO operator. Taking the Fourier transform of both sides of Equation 1.1, the RAO is defined as the ratio of incident wave forcing to buoy response, given by:

$$\mathbf{RAO}(\omega) = \frac{\underline{\underline{\Xi}}(\omega)}{A_w} = [\underline{\underline{C}} - \omega^2(\underline{\underline{M}} + \underline{\underline{A}}(\omega)) - i\omega\underline{\underline{B}}(\omega)]^{-1} \frac{\mathbf{f}_{\text{exc}}(\omega)}{A_w} \quad (1.6)$$

This fraction tells us how much the perceived wave (ie buoy displacement) is amplified or damped by the buoy/floe geometry. Note that this function changes as the floe the buoy is situated on changes, which is why wave statistics from buoy dynamics could be misleading if the buoy-ice regime changes or isn't fully quantified. Previously, a RAO of 1 was assumed, but as ice floe size increases the RAO decreases (more dispersion/scattering).

1.3 Signal Processing and Instrumentation

The signal processing pipeline used by [?] follows the following broad steps to calculate significant wave height and mean period. A very similar method is also described in [?] and [?].

Recall that the energy of a signal $y(t)$ is given by:

$$E = \int_{-\infty}^{\infty} |y(t)|^2 dt = \int_{-\infty}^{\infty} |Y(f)|^2 df \quad (1.7)$$

And the power spectral density (PSD) can be shown to be (through autocorrelation and the Weiner Khinchine theorem):

$$\hat{P}(f) = \frac{1}{T} |Y(f)|^2 \quad (1.8)$$

The two-sided periodogram estimate of the PSD of a signal y , $\hat{P}_{yy}$, of length N with K zero padding is given by:

$$\hat{P}_{yy} = \frac{1}{(N + K)\Delta t} |Y_k|^2 \quad (1.9)$$

where Y_k is the Discrete Fourier Transform (DFT) of signal $y(t)$.

Note that the division by the sample period Δt turns the energy spectrum into a power spectrum. Given that the random surface elevation time series is real-valued, the power spectrum is symmetrical, and so a single-sided PSD estimate is given by:

$$\text{PSD}(f) = 2\hat{P}_{yy} = \frac{2}{(N + K)\Delta t} |Y_k|^2 \quad (1.10)$$

and with proper scaling can be shown to be equivalent to the energy spectrum $E(f)$.

1.3.1 The Welch Method

Truncating the measured signal, along with a finite sampling frequency causes the variance of the recovered PSD to be approximately the square of the variance of the true PSD, regardless of the length of the original truncated time series. To deal with this, the variance is reduced by averaging several periodograms at the cost of spectral resolution. Welch's method assists by windowing the signal by a smooth window function of each periodogram and by overlapping these periodograms to increase the number of periodograms taken of a time series. There needs to be a compensating factor introduced as windowing reduces power.

The averaged periodogram estimate $\hat{P}(f_n)$ across K segments is given by:

$$\hat{P}(f_n) = \frac{1}{K} \sum_{k=1}^K I_k(f_n) \quad (1.11)$$

where $I_k(f_n)$ is the k -th modified periodogram estimate given by:

$$I_k(f_n) = \frac{L}{U} |A_k(n)|^2 \quad (1.12)$$

where $A_k(n)$ is the Discrete Fourier Transform (DFT) of the k -th windowed data segment X_k of length L , given by:

$$A_k(n) = \frac{1}{L} \sum_{j=0}^{L-1} X_k(j) W(j) e^{-i2\pi jn/L} \quad (1.13)$$

and U is a normalization factor for power given by:

$$U = \frac{1}{L} \sum_{j=0}^{L-1} W^2(j) \quad (1.14)$$

The important result is that the key parameters of interest are derived from the energy spectrum by taking its spectral moments $m_n = \int_0^\infty f^n E(f) df$.

Table 1.1: Ocean Wave Spectral Parameters

Parameter	Equation	Description
T_{m01} or $\bar{T}$	$\frac{m_0}{m_1}$	The mean wave period.
T_{m02} or T_2	$\sqrt{\frac{m_0}{m_2}}$	The average up-crossing period between waves.
H_{m0} or H_s	$4\sqrt{m_0}$	The significant wave height.

Note that other signal processing endeavors are taken, such as filtering, data fusion (Kalman filtering), accounting for IMU displacement off the bottom of buoy (height) and IMU noise characteristics.

1.4 Flexural Response of Intermediate Floes

The theory of flexural responses of ice floes to wave forcing has been of keen interest to scientists and engineers. In 1994, Meylan and Squire reported a precise, linear model of a solitary ice floe in ocean waves, allowing the floe to bend with the passing wave [?]. The theory was strictly restricted to 2D analysis, where the floe cases had variable diameter (length) and constant thickness. The same theory was expanded in [?] for the case of 3D circular disks. Finally in 2002, Meylan and Squire were able to model the wave response of floes of **arbitrary** geometry, and their findings were validated with wave tank experiments in 2015 [?]. The methods discussed in these two papers are what I believe provide the most actionable insight to the devised pipeline.

1.5 3D response of floes of arbitrary geometry: Theory

The main theory underlining this section is found in [?], but I will attempt to summarize their process. Where there are elements of my understanding that are missing, I will highlight for your attention.

This paper outlines the full 3D model for the motion and bending of a solitary ice floe of arbitrary (this word is maybe abused here, due to the low computational ability and time restraints they tested only a few geometries) geometry. A key assumption of theirs is that the floe has a **shallow draft**, ie can be modeled as a thin plate. This is quite important, because stiffness is obviously very dependent on thickness, along with aspect ratio (width:length).

The overarching idea is to express the vertical displacement of the floe in its vibration modes, and substitute them into the integral equation of motion, providing a linear system whose coefficients reveal the ice motion. A notable conclusion, one which is potentially utilizable in the pipeline, is that stiffness is a very important factor with respect to motion and energy scattering, and that above a **critical stiffness**, the geometry also has a large impact.

Their system applies to the intermediate floe regime, and they propose that the regimes can be differentiated by the relative size of floe diameter to incident wavelength:

1. Small floes (rigid body): $D \ll \lambda$
2. Intermediate floes (flexural + rigid body): $D \approx \lambda$
3. Large consolidated sheet (infinite and flexible): $D \gg \lambda$

1.5.1 Equation of motion of the floe

Let the floe be modeled as a thin plate of constant thickness and shallow draft. The thin plate equation given by Hildebrand [?] is:

$$D\nabla^4 W + \rho_i h \frac{\partial^2 W}{\partial t^2} = p \quad (1.15)$$

Where W is the floe displacement, p is the pressure on the underside of the floe and D is the **flexural rigidity** defined as:

$$D = \frac{Eh^3}{12(1 - \nu^2)}$$

with E being Young's modulus, h being the thickness of the floe, and ν being Poisson's ratio. Note that D is dependent on the cube of the thickness, hence increasing the thickness dramatically increases the resistance to wave bending relative to Young's modulus.

The first term in the thin plate equation represents the flexural resistance (recall that the biharmonic operator ∇^4 calculates spatial bending in 2D). The second term captures the physical inertia of the moving floe (mass/unit area multiplied by acceleration). Assuming the plate is in contact with the water at all times, and that equation 1.15 is (obviously) subject to the free edge boundary conditions for a thin plate:

$$\frac{\partial^2 W}{\partial n^2} + \nu \frac{\partial^2 W}{\partial s^2} = 0 \quad \text{and} \quad \frac{\partial^3 W}{\partial n^3} + (2 - \nu) \frac{\partial^3 W}{\partial n \partial s^2} = 0$$

Where n and s respectively represent the normal and tangential directions, the above equation simply captures the zero bending moment and zero shear forces along the boundary. Finally, the pressure is subject to the linearized Bernoulli equation at the water surface, namely that the pressure on the underside of the floe is equal to the sum of the unsteady dynamic pressure (hydrodynamic force imparted onto ice by wave motion in the fluid potential field Φ) and the hydrostatic restoring pressure (buoyancy force):

$$p = -\rho \frac{\partial \Phi}{\partial t} - \rho g W \quad (1.16)$$

Meylan proceeds to nondimensionalize the length variables with respect to a where the surface area of the floe is $4a^2$. The time variables are nondimensionalized with respect to $\sqrt{g/a}$ and the mass variables with respect to ρa^3 .

$$\bar{x} = \frac{x}{a}, \quad \bar{y} = \frac{y}{a}, \quad \bar{z} = \frac{z}{a}, \quad \bar{W} = \frac{W}{a}, \quad \bar{t} = t\sqrt{\frac{g}{a}}, \quad \bar{\Phi} = \frac{\Phi}{a\sqrt{ga}}$$

Note that because it is a luss I will omit the overbars from now onwards, and it will be implied. Substituting these values back into the governing equations, equations 1.15 and 1.16 become:

$$\beta \nabla^4 W + \gamma \frac{\partial^2 W}{\partial t^2} = \frac{\partial \Phi}{\partial t} - W \quad (1.17)$$

Where the **dimensionless stiffness parameter** β and **dimensionless mass parameter** γ are represented as:

$$\beta = \frac{D}{\rho g a^4} \quad \text{and} \quad \gamma = \frac{\rho_i h}{\rho a}$$

Considering the response to wave forcing of a single frequency (although more complex responses can be constructed via superposition), and that the equations of motion are linear, then the displacement must have the same frequency as the incoming wave. Letting α be the nondimensionalized wave number, the time-harmonic representations of the displacement and fluid velocity potential are:

$$\bar{W}(x, y, t) = \text{Re} \left[w(x, y) e^{-i\sqrt{\alpha}t} \right]$$

and

$$\bar{\Phi}(x, y, z, t) = \text{Re} \left[\phi(x, y, z) e^{-i\sqrt{\alpha}t} \right]$$

Plugging these into Equation 1.17 yields:

$$\beta \nabla^4 w + \alpha \gamma w = \sqrt{\alpha} - w \quad (1.18)$$

Here $w(x, y)$ is the frequency domain spatial elevation pattern (as we have split up time dependence). Don't worry, we're almost done. Just need to resolve the water motion, then combine!

1.5.1.1 Equations of motion of the water

Utilizing the nondimensionalized variables, and assuming linear potential flow theory: ie the fluid is inviscid and irrotational, the boundary value problem describing linear surface gravity waves is [?]:

$$\left. \begin{aligned} \nabla^2 \phi &= 0, & -\infty < z < 0, \\ \partial \phi / \partial z &= 0, & z \rightarrow -\infty, \\ \partial \phi / \partial z &= -i\sqrt{\alpha} w, & z = 0, \mathbf{x} \in \Delta, \\ \partial \phi / \partial z - \alpha \phi &= 0, & z = 0, \mathbf{x} \notin \Delta \end{aligned} \right\} \quad (1.19)$$

The variable $\mathbf{x} = (x, y)$ is a point on the water surface, and Δ is the region of the water surface occupied by the floe. The above boundary value problem is subject to the Sommerfield radiation condition; at large distances the potential consists of a radial outgoing wave (generated by floe motion) and the incident wave, expressed as:

$$\lim_{|\mathbf{x}| \rightarrow \infty} \sqrt{|\mathbf{x}|} \left(\frac{\partial}{\partial |\mathbf{x}|} - i\alpha \right) (\phi - \phi_{in}) = 0 \quad (1.20)$$

And the incident potential ϕ_{in} is:

$$\phi_{in}(x, y, z) = \frac{A}{\sqrt{\alpha}} e^{i\alpha(x \cos \theta + y \sin \theta)} e^{\alpha z} \quad (1.21)$$

Where A is the nondimensionalized wave amplitude. Now comes the process I do not fully understand. A standard process to solving linear wave problems is by condensing the boundary value problem into an integral equation through the incorporation of a Green's function. The idea is to reduce the 3D fluid potential BVP into a 2D surface integral equation over the floe's wetted domain Δ . The formulation/choice of Green's functions is explained in more detail in [?].

The Green's function integral equation collapses equations 1.19 and 1.20 to:

$$\phi(\mathbf{x}) = \phi_{in}(\mathbf{x}) + \iint_{\Delta} G_{\alpha}(\mathbf{x}, \mathbf{y}) [\alpha \phi(\mathbf{y}) + i\sqrt{\alpha} w(\mathbf{y})] dS_{\mathbf{y}} \quad (1.22)$$

From my understanding, this equation can be understood as follows:

1. $\phi_{in}(\mathbf{x})$ is obviously the incoming wave passing through (as if floe is not present)
2. $[\alpha \phi + i\sqrt{\alpha} w]$ is the point source strength at position $\mathbf{y}$. The vibrating ice floe acts like a continuous sheet of tiny wave generators, and the motion strength at position $\mathbf{y}$ depends on the ice vertical displacement (w) and the dynamic water pressure ϕ .
3. $\iint_{\Delta} G_{\alpha}(\mathbf{x}, \mathbf{y}) [\dots] dS_{\mathbf{y}}$ can be understood to be the sum of (convolution) the ripple effects originating from every single tiny patch at $\mathbf{y}$ to find the total scattered wave field at location $\mathbf{x}$.

The Green's functions that encapsulate the BVPs are postulated to have the form:

$$G_{\alpha}(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi} \left(\frac{2}{|\mathbf{x} - \mathbf{y}|} - \pi\alpha [H_0(\alpha|\mathbf{x} - \mathbf{y}|) + Y_0(\alpha|\mathbf{x} - \mathbf{y}|)] + 2\pi i\alpha J_0(\alpha|\mathbf{x} - \mathbf{y}|) \right) \quad (1.23)$$

where J_0 and Y_0 are Bessel functions of the 1st and 2nd kind respectively, of order zero. H_0 is the Struve function of order zero. I have very little notion as to why this specific Green's function satisfies the conditions, and would appreciate a little help here (only if you believe it necessary...).

The integral equation 1.22 can be solved numerically.

1.5.2 Solving for Wave-Induced Motion

To solve for ice floe motion, equations 1.18 and 1.22 must be solved simultaneously. To do so, Meylan expands the floe motion in the free modes of vibration of a thin plate, where only a

select few geometries have analytical solutions. More complex shapes require numerical solutions through finite element methods (FEM).

Now for some complicated stuff. Because ∇^4 is **self-adjoint** (subject to the free edge boundary conditions), the free vibration shapes of an ice floe form a set of basis functions such that:

$$w(\mathbf{x}) = \sum_{i=1}^N c_i w_i(\mathbf{x}) \quad (1.24)$$

where each mode w_i is orthogonal, and has eigenvalue λ_i being positive and real (spectral theorem). Moreover, assuming they are orthonormal allows for the condition that the inner product of any 2 modes (surface integral of the product of the 2 modes over some subset of the wetted surface) must equal the Kronecker delta. The eigenvalues approach infinity as their index does the same, and they chose to order the modes by increasing eigenvalue.

The expand the displacement of the floe in a finite number of modes (N):

$$w(\mathbf{x}) = \sum_{i=1}^N c_i w_i(\mathbf{x}) \quad (1.25)$$

and by the linearity of 1.22, the potential can be written as

$$\phi(\mathbf{x}) = \phi_0(\mathbf{x}) + \sum_{i=1}^N c_i \phi_i(\mathbf{x}) \quad (1.26)$$

where ϕ_0 satisfies the integral equation (diffraction potential ie rigid/fixed floe)

$$\phi_0(\mathbf{x}) = \phi_{\text{in}}(\mathbf{x}) + \iint_{\Delta} \alpha G_{\alpha}(\mathbf{x}, \mathbf{y}) \phi_0(\mathbf{y}) dS_{\mathbf{y}} \quad (1.27)$$

and ϕ_i satisfies (radiation potential, ie vibrating mode 'i'):

$$\phi_i(\mathbf{x}) = \iint_{\Delta} G_{\alpha}(\mathbf{x}, \mathbf{y}) [\alpha \phi_i(\mathbf{y}) + i\sqrt{\alpha} w_i(\mathbf{y})] dS_{\mathbf{y}} \quad (1.28)$$

Subbing these into equation 1.18 yields:

$$\beta \sum_{i=1}^N \lambda_i c_i w_i - \alpha \gamma \sum_{i=1}^N c_i w_i = -i\sqrt{\alpha} \left(\phi_0 + \sum_{i=1}^N c_i \phi_i \right) - \sum_{i=1}^N c_i w_i \quad (1.29)$$

and finally the continuous PDE is reduced to a dense NxN matrix system with unknown $\mathbf{c} = [c_1, c_2, \dots, c_N]^T$ containing the modal response amplitudes:

$$(\beta \lambda_j + 1 - \alpha \gamma) c_j = -i\sqrt{\alpha} \iint_{\Delta} \left(\phi_0(\mathbf{x}) + \sum_{i=1}^N c_i \phi_i(\mathbf{x}) \right) w_j(\mathbf{x}) dS_{\mathbf{x}} \quad (1.30)$$

Note that the last 2 equations is essential a **Galerkin projection**. Moreover, it has been ordered around to have the form $\underline{\underline{\mathbf{A}}} \mathbf{c} = \mathbf{b}$.

1.5.3 Conclusion/Summary

This paper provides a coupled boundary element and finite element framework that models the three-dimensional flexural motion, wave scattering, and wave-induced drift forces for flexible ice floes of arbitrary geometry. By solving the velocity potential in the fluid domain using surface Green's functions and representing the ice plate deflections through modal expansion, the model calculates the full spatial displacement field, scattered wave energy via the Kochin function, and second-order time-averaged drift forces and yaw moments. A central finding of the paper is that floe response, wave scattering, and drift forces are governed overwhelmingly by non-dimensional flexural rigidity rather than structural mass. Furthermore, the paper identifies a critical stiffness threshold below which floe geometry has a negligible impact on scattering, allowing simple circular approximations, and above which the specific geometry significantly alters the response.

This theoretical foundation is directly relevant as we develop data-driven approaches, such as Dynamic Mode Decomposition (DMD), to model and analyze ice floe behavior. Because Meylan's formulation explicitly decomposes physical floe deflection into a linear combination of spatial vibration modes oscillating at distinct frequencies, it provides a solid physical baseline that aligns naturally with what data-driven spatial-temporal decomposition methods aim to extract from observational data.

The paper also offers practical insight for physical sensing and data collection by demonstrating that vertical displacement and acceleration vary much more smoothly across the floe surface than internal bending strain. This provides a strong physical justification for using acceleration or motion sensors like inertial measurement units, as they capture a far cleaner signal for data-driven algorithms than strain gauges. Finally, the strong sensitivity of the motion to flexural stiffness indicates that inferring physical properties like ice thickness from observed motion is a mathematically well-posed problem, while the concept of a critical stiffness threshold gives us a clear rule of thumb for when simplified floe shapes can be used in our computational framework instead of complex two-dimensional spatial geometries.

1.5.4 Intermediate floes: Validation

[?] takes the arbitrary geometry theory from the previous section and validates it against wave tank data from a floe that was free to surge, drift, and overwash (unlike some earlier constrained validation setups), so it's a genuine test of the linear theory under fairly messy, realistic conditions. I'll skip most of the experimental description and focus on the modal decomposition and the resulting matrix system, since that's what's relevant to us. The basic setup was: floe shapes of rectangular geometry of fixed length and width, but tested with 2 different thicknesses and plastic (hence density and Young's modulus variability). Markers were placed on the surface of each floe, and a camera was used to track the time series displacements of each.

1.5.4.1 Modal Decomposition of the Measured Response

Rather than working with raw marker displacements directly, the measured surface motion is projected onto the floe's known in-vacuo natural vibration modes $w_j(\mathbf{x})$ ie the same eigenfunctions of the biharmonic operator from the previous section:

$$\nabla^4 w_j = \mu_j^4 w_j \quad \text{for } \mathbf{x} \in \Omega, \text{ subject to free-edge conditions,} \quad (1.31)$$

where μ_j are the eigenvalues, computed once via FEM for the known floe geometry and independent of the forcing.

For each marker m , after transients decay, the vertical displacement is fit to a single-frequency harmonic at the (known) incident wave frequency ω via least-squares, giving a complex amplitude A_m such that $\eta_m(t) \approx \text{Re}\{A_m e^{-i\omega t}\}$. These complex amplitudes are then jointly projected onto a chosen finite subset of modes Λ :

$$A_m \approx \sum_{j \in \Lambda} \xi_j^{\text{ex}} w_j(\mathbf{x}_m), \quad m = 1, \dots, 16 \quad (1.32)$$

i.e. a second least-squares problem, now solving for the modal weights ξ_j^{ex} given a known mode basis and known sensor locations, rather than solving for the spatial field itself. $\Lambda = \{1, 2, 5, 6, 7, 9, 11\}$ was used; where the first two are the rigid body heave and pitch modes, the remaining five the dominant flexural modes, restricted to those symmetric about the incident wave direction.

1.5.4.2 The Governing Matrix System

The theoretical side is solved with the same modal expansion as [?], but written out here in an explicit, directly reusable matrix form:

$$(\underline{\underline{K}} + \underline{\underline{C}} - \omega^2 \underline{\underline{M}} - \omega^2 \underline{\underline{A}}(\omega) - i\omega \underline{\underline{B}}(\omega)) \underline{\underline{\xi}}^{\text{th}} = \underline{\underline{f}}(\omega) \quad (1.33)$$

with

$$\underline{\underline{K}} = [\beta \mu_j^4], \quad \underline{\underline{C}} = \underline{\underline{I}}, \quad \underline{\underline{M}} = \gamma \underline{\underline{I}}, \quad (1.34)$$

where $[\cdot]$ denotes a diagonal matrix, β and γ are the same nondimensional stiffness and mass parameters from Section 3, and $\underline{\underline{A}}(\omega), \underline{\underline{B}}(\omega)$ are frequency-dependent modal added mass and damping matrices computed via the same Green's-function boundary integral as before, now taking inner products against the mode shapes w_i, w_j rather than raw panel basis functions.

This is structurally identical to the rigid body Cummins equation in Section 2: same mass/damping/stiffness/added-mass template, just projected onto the flexural mode basis instead of 6 rigid DOFs, with $\underline{\underline{A}}(\omega), \underline{\underline{B}}(\omega)$ frequency-dependent in both cases. I think this unifying structure is worth stating explicitly somewhere in the report: all three regimes appear to be instances of the same generalized damped oscillator equation, differing only in modal basis and in how β, γ scale.

1.5.4.3 Relevance to the Pipeline

The main result: that the linear modal model matches the measured modal weights well even under nonlinear surface contamination is reassuring for the SSI/ERA framing: the flexural response behaves as a linear(ish), non-autonomous system driven by the wave field, and this holds up under realistic rather than idealized forcing.

The more useful point for Phase 2/3 discussion, though, is the contrast in problem conditioning. Equation 1.32 is a well-posed least-squares problem: 16 sensors, known geometry, known mode basis, solving for 7 modal weights. Our problem is the underdetermined inverse of this: one sensor, unknown geometry, and we additionally want to recover the physical parameters (β, γ , and ultimately R, D, E^v) that determine the mode basis itself, not just the weights on a known basis. If 16 sensors were needed to comfortably resolve 7 modes here, that's a useful calibration point for how many modes/parameters we can realistically expect to identify from a single IMU.

Equation 1.33 is also directly usable as a forward model for that twin test: solve for $\underline{\underline{\xi}}^{\text{th}}$ at a given ω , evaluate $w(\mathbf{x}_{\text{sensor}}) = \sum_j \xi_j^{\text{th}} w_j(\mathbf{x}_{\text{sensor}})$ at the IMU location, and that's a synthetic single-point response with known ground-truth β, γ to test identifiability against.

1.5.5 Montiel 2013: Circular floes

A much less computationally costly approach is to use the closed form analytic solutions derived by [?], and their experimental validation [?]. Note that they only modelled **circular disc** floes, unlike the arbitrary geometries discussed earlier (which required FEM).

They utilize similar assumptions as [?] (except the obvious geometry restriction), and instead of solving the BVP via Green’s functions, they utilize an eigenfunction matching method (EMM).

The disc of thickness D and radius R floats on fluid of finite depth h , in cylindrical coordinates (r, θ, z) . The draught follows Archimedes’ principle, $d = (\rho/\rho_0)D$. The governing equations are the same Laplace/free-surface/thin-plate system as before, but now written for the disc-covered region ($r < R$) versus the open water region ($r > R$) separately:

$$\partial_z \phi = \alpha \phi \quad (r > R, z = 0), \quad (\beta \nabla^4 + 1 - \alpha d) \partial_z \phi = -\alpha \phi \quad (z = -d) \quad (1.35)$$

The first condition applies on the open water surface outside the disc: it says the vertical motion of the surface must match what the local wave pressure “tells it” to do, with α setting the proportionality. This is the standard open-ocean surface condition – the disc plays no role in it yet.

The second condition applies underneath the disc instead. It’s the same idea, but now some of that vertical motion gets “used up” bending the disc rather than just displacing water – that’s what the $\beta \nabla^4$ term captures: how much bending resistance there is (via β) times how curved the disc currently is (via ∇^4). If $\beta = 0$ (an infinitely floppy disc, or equivalently no disc at all) this term disappears and we’re back to a free-surface-like condition. If $\beta \rightarrow \infty$ (an infinitely rigid disc), bending is suppressed entirely and only rigid heave/pitch motion remains.

The wave field outside the disc, $\phi^{(o)}$, is split into two pieces:

$$\phi^{(o)} = \phi_I + \phi_S \quad (1.36)$$

ϕ_I is the incoming wave exactly as it would look with no disc present at all; fully known, prescribed at the start. ϕ_S is the *scattered* wave: the extra disturbance created because the disc is actually there, blocking and reflecting the incoming wave. This is the unknown part we need to solve for.

Far away from the disc, the scattered wave has to behave sensibly – it should look like an outward-travelling wave that doesn’t come back in from infinity. This physical requirement (the Sommerfeld radiation condition) is what allows the problem to be solved uniquely, rather than admitting spurious solutions where energy appears from nowhere at the boundary.

Rather than solving directly for ϕ_S as some arbitrary function, the method expresses it as a sum of simple building-block functions, each of which already satisfies the depth and radiation conditions by construction:

$$\phi_S(x) \approx \sum_{m=0}^M \varphi_m^{(o)}(z) \sum_{n=-N}^N A_{m,n}^{(o)} K_n(k_m r) e^{in\theta} \quad (1.37)$$

Breaking this down piece by piece:

- $e^{in\theta}$: describes how the wave varies going around the disc angularly. n is just an integer counting how many angular “lobes” that piece has.
- $K_n(k_m r)$: describes how the wave decays with distance r from the disc. K_n is a known special function (a modified Bessel function), so this isn’t something we solve for – it’s a fixed shape, we just don’t yet know how much of it to include.

- $\varphi_m^{(o)}(z) = \cos k_m(z + h)$: describes how the wave varies with depth z . The allowed values of k_m aren't free either; they're fixed by requiring the whole thing to also satisfy the surface condition, which forces k_m to be roots of:

$$k \tanh(kh) = -\alpha \quad (1.38)$$

This equation has one purely imaginary root k_0 (this is the ordinary travelling wave you'd see on the open ocean) and infinitely many real roots $k_m, m \geq 1$ (these don't travel; they decay exponentially away from the disc edge and only matter very close to it).

- $A_{m,n}^{(o)}$: the one genuinely unknown quantity in the whole expression: a number telling us how much of each building-block piece is actually present. This is what we ultimately solve for. Essentially the "weights".

So the whole approach is: pick out the fixed set of "legal" shapes the water motion is allowed to take (fixed by physics and geometry), then solve a linear problem for how much of each shape is needed to match the disc's actual boundary conditions. In practice only a handful of the real (k_m , evanescent) modes matter, since their contribution shrinks rapidly as m increases – which is why the sum can be safely truncated at some finite M .

The same idea is applied underneath the disc, giving a second expansion for $\phi^{(i)}$ (the potential in the disc-covered region), except now the allowed depth-shapes are governed by a modified dispersion relation that includes the disc's stiffness:

$$(\beta\kappa^4 + 1 - \alpha d) \kappa \tan \kappa(h - d) = -\alpha \quad (1.39)$$

This has the same purely-imaginary-plus-real-roots structure as the open water case (κ_0 imaginary, $\kappa_p, p \geq 1$ real), **but also two extra roots, κ_{-2} and κ_{-1} , which are complex**. These only exist because of the $\beta\kappa^4$ term, they have no counterpart in the open water problem, and physically represent a travelling wave under the disc that also loses energy as it goes (a damped travelling wave), which makes sense: energy is continuously being fed into bending the disc as the wave passes underneath it.

Finally, matching the two solutions (open water and disc-covered) at the boundary $r = R$; requiring the pressure and vertical velocity to agree there, turns everything into a finite system of linear equations for the unknown amplitudes $A_{m,n}^{(o)}$ and $A_{p,n}^{(i)}$, which is then just solved numerically.

Once the water problem is solved, the resulting disc deflection can be re-expressed in a more physically intuitive basis: the disc's own natural modes of vibration $w_{n,j}(r, \theta)$ (the shapes it would vibrate in if you just tapped it, with no water present at all), indexed by n (number of nodal diameters, lines across the disc that don't move) and j (number of nodal circles, concentric rings that don't move):

$$w(r, \theta) = \sum_{n,j} A_{n,j} w_{n,j}(r, \theta) \quad (1.40)$$

The lowest two modes are just heave ($w_{0,0}$, the whole disc moving up and down rigidly) and pitch/roll ($w_{\pm 1,0}$, the whole disc rocking rigidly), these carry no actual bending. Everything beyond that is genuine flexure. In practice only a small number of the lowest modes end up mattering for realistically-sized floes and wavelengths, echoing the same point from the Meylan 2015 summary.

The purely elastic version of this model (constant, real E) consistently *overpredicted* how much the disc bends compared to what's measured experimentally: by a lot for the flexural modes specifically, while the rigid heave/pitch modes are matched well. This points to some damping mechanism, specific to bending, that a purely elastic (lossless) material can't capture.

The fix: instead of treating E as a fixed real number, treat it as complex and frequency-dependent, $E^v(\omega)$, built from a simple three-parameter spring/dashpot model of the material (a spring in series with a spring-and-damper pair). Under steady oscillation this reduces to:

$$E^v(\omega) = \frac{q_0 + i\omega q_1}{1 + i\omega p_1}(1 + \nu) \quad (1.41)$$

where p_1, q_0, q_1 are just fixed combinations of the three spring/dashpot constants. The imaginary part of $E^v(\omega)$ is what represents energy loss (damping) – a purely real E^v recovers the original lossless elastic model exactly.

The nice part is that nothing else about the model needs to change: β just becomes complex and frequency-dependent too, $\beta^v(\omega) = E^v(\omega)D^3/12\rho_0g(1 - \nu^2)$, and Equation 1.39's roots κ_p become complex roots $\kappa_p^v(\omega)$ – close to, and paired one-to-one with, the original real/imaginary roots, so the original roots can be used as a good starting guess when numerically solving for the new ones.

Why is this paper relevant? Well: (R, D, E^v) disc radius, thickness, and complex viscoelastic modulus – is exactly our target parameter set for the inversion, and here it falls out naturally as the fix needed to explain real damping in an experimental setting very close to ours (a floating elastic plate, point-wise displacement measurements, regular wave forcing).

Two things worth flagging:

1. **Equation 1.39 (viscoelastic version) looks like the concrete object Phase 3 needs to fit against.** Rather than the temporal-to-spatial eigenvalue mapping being something abstract to invent, the task looks like: given poles from Phase 2, find (R, D, E^v) so that the corresponding roots $\kappa_p^v(\omega)$ of Equation 1.39 match them. That reframes Phase 3 as fitting parameters into an existing, known relation, not deriving new theory.
2. **The extra complex roots existing only because of the $\beta\kappa^4$ term** gives a cheap sanity check: a synthetic rigid-body-only case ($\beta \rightarrow \infty$ or $\beta = 0$) should make those roots collapse back to the ordinary open-water structure. That's an easy correctness check to run on the twin-test code before trusting it in the flexure-dominated regime.

1.6 Data-Driven Methods

Most of what follows was learned from Nathan Kutz's AMATH563 course from UW.

1.6.1 Model Discovery

The main idea here is outlined in brief in [?]. It utilized **sparse regression** to find the optimal set of functions from a set library which satisfies the underlying set of ODEs. It is a model discovery algorithm designed to find underlying physical laws, purely using data-driven methods.

Consider a matrix system of ODEs given by:

$$\dot{\mathbf{u}} = \mathbf{f}(\mathbf{u}, \mu(t))$$

where $\mathbf{u} \in \mathbb{R}^n$ represents the state variables, $\mu(t)$ accounts for explicit parameter variation over time, and $\mathbf{f}(\cdot)$ is sparse (composed of only a few active nonlinear functions).

Time-series measurements $\mathbf{u}$ are collected, and their derivatives calculated (incredibly important that the numerical differentiation scheme has high accuracy!) and are organized into data matrices

$\mathbf{X}, \dot{\mathbf{X}} \in \mathbb{R}^{m \times n}$ across m time steps. A candidate library $\Theta(\mathbf{X}) \in \mathbb{R}^{m \times M}$ containing M potential right-hand side functions is constructed:

$$\Theta(\mathbf{X}) = [\mathbf{g}_1(\mathbf{u}) \quad \mathbf{g}_2(\mathbf{u}) \quad \mathbf{g}_3(\mathbf{u}) \quad \dots \quad \mathbf{g}_M(\mathbf{u})]$$

The governing equations are formulated as a linear matrix problem:

$$\dot{\mathbf{X}} = \Theta \Xi$$

where $\Xi \in \mathbb{R}^{M \times n}$ is a coefficient matrix determining the active terms in $\mathbf{f}$.

Standard unconstrained least-squares regression yields a dense weight matrix Ξ , producing overly complex models that overfit noise. Promoting parsimony requires sparse optimization: Ideal ℓ_0 -Regularization (NP-Hard):

$$\hat{\Xi} = \arg \min_{\Xi} \|\Theta \Xi - \dot{\mathbf{X}}\|_2^2 + \lambda \|\Xi\|_0$$

This directly minimizes non-zero terms but is combinatorially intractable.

ℓ_1 -Convex Relaxation: Replaces $\|\Xi\|_0$ with an ℓ_1 penalty $\|\Xi\|_1$ (e.g., LASSO, Elastic Net), transforming the problem into a tractable convex optimization task. Sequential Least-Squares Thresholding: A computationally efficient alternative that initially minimizes an ℓ_2 ridge loss:

$$\hat{\Xi} = \arg \min_{\Xi} \|\Theta \Xi - \dot{\mathbf{X}}\|_2^2 + \lambda \|\Xi\|_2^2$$

After solving, coefficients smaller than a defined threshold λ are set to zero. The linear system is iteratively re-solved over the remaining non-zero terms until the active coefficient matrix Ξ converges.

This methodology is something I have actually interacted with, writing a MATLAB script to reconstruct the Van der Pol oscillator. Noise effects have also been investigated.

A very similar process is used to reconstruct PDEs. A brief aside is that time delayed embeddings can be used for model discovery through the construction of the **Hankel matrix**. This method is used when you have multiple state variables but can only measure an incomplete set of them. Running time delay embeddings will highlight the true order of the dynamics (again something I have written before).

1.6.2 Dynamic Mode Decomposition

Similar to SINDy, except now you enforce a linear system. In fact the whole goal is to find the best fit linear system that fits the dynamics. Given a dynamical system along with some measurements of the state variables, write the variable $\mathbf{x}$ as a linear combination of eigenfunctions of the linear system $A\mathbf{x} = \mathbf{b}$. Sorry for the lack of underlines, the latex editor is messing with my maths notation so I will continue writing in this paragraph until it goes onto the next page wow this is taking much longer than I expected, I wonder if it will ever end.

Lol

Lol

Lol

Sorry about that. Back to business. Let the dynamical process be defined by:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, t, \mu) \tag{1.42}$$

And the measurements being $\mathbf{y}_k = g(\mathbf{x})$ where $\mathbf{x} \in \mathbb{R}^n$ for a large n (ie $\mathbf{x}$ is super tall and skinny). For $\frac{d\mathbf{x}}{dt} = \underline{\underline{A}}\mathbf{x}$, write $\mathbf{x}$ as a linear combination of the system's eigenfunctions:

$$x = \sum_{j=1}^n b_j \phi_j e^{\lambda_j t} \quad (1.43)$$

The issue is getting the matrix $\underline{\underline{A}}$. Assuming you can measure $\mathbf{x}$, and stacking each time snapshot as columns of the matrix $\underline{\underline{X}}$ along with creating a time delayed matrix $\underline{\underline{X'}}$ ie the same as the previous but with col 1 being x_2 ie $\mathbf{x}$ at $t = 2$ instead of $\mathbf{x}$ at $t = 1$. Note the matrix is the same size, hence has last column being $\mathbf{x}$ at $t = m$ instead of $m-1$.

$\underline{\underline{X'}}$ is simply the next step in the dynamical system (per row). Thus the idea is to find the matrix $\underline{\underline{A}}$ such that:

$$\underline{\underline{X'}} = \underline{\underline{A}}\underline{\underline{X}} \quad (1.44)$$

Thus you can just recover $\underline{\underline{A}}$ by the pseudo-inverse of $\underline{\underline{X}}$, ie $\underline{\underline{A}} = \underline{\underline{X'}}\underline{\underline{X}}^\dagger$

The issue is that $\mathbf{x}$ is large, thus the pseudo-inverse is massive with a huge outer product. To reduce this, use SVD approximations for $\underline{\underline{X}}$ ie only keeping the most dominant singular values. This is done through rank r truncation $\underline{\underline{X}} = \underline{\underline{U}}\underline{\underline{\Sigma}}\underline{\underline{V}}^*$. Plug this into the expression of $\underline{\underline{X'}}$ for $\underline{\underline{X}}$ to yield:

$$\underline{\underline{A}} = \underline{\underline{X'}}\underline{\underline{V}}\underline{\underline{\Sigma}}^{-1}\underline{\underline{U}}^* \quad (1.45)$$

Rebuild the dynamics in the rxr subspace $\tilde{\underline{\underline{A}}} = \underline{\underline{U}}^*\underline{\underline{A}}\underline{\underline{U}} = \underline{\underline{U}}^*\underline{\underline{X'}}\underline{\underline{V}}\underline{\underline{\Sigma}}^{-1}$. Then solve the eigenvalue problem $\tilde{\underline{\underline{A}}}\underline{\underline{W}} = \underline{\underline{W}}\underline{\underline{\Lambda}}$. Here $\underline{\underline{\Lambda}}$ has diagonal $= \lambda_j$ with size rxr. The DMD modes can thus be calculated by going back to nxn space:

$$\underline{\underline{\Phi}} = \underline{\underline{X'}}\underline{\underline{V}}\underline{\underline{\Sigma}}^{-1}\underline{\underline{W}} \quad (1.46)$$

Thus allowing us to write $\mathbf{x}(t) = \underline{\underline{\Phi}}e^{\underline{\underline{\Lambda}}t}\underline{\underline{b}} = \sum_{k=1}^r \phi_k e^{\omega_k t} b_k$ and IC: $\mathbf{x}(0) = \underline{\underline{\Phi}}\underline{\underline{b}}$; $\underline{\underline{b}} = \underline{\underline{\Phi}}^{\backslash} \mathbf{x}(0)$

This is a lot. I have also gone through Koopman theory, but I will leave that for next time.

1.7 The Inversion Pipeline: Three Phases

The overall goal, as stated in the introduction, is to invert single-sensor IMU response data to recover hydroelastic properties of the floe it's sitting on. I'm breaking this into three phases:

1. **Forward modelling.** Given a floe geometry and a set of hydroelastic parameters (thickness, stiffness, viscoelastic properties), predict the floe's response to a given incident wave field. This is the well-posed, "solved" direction of the problem, handled by existing theory: Montiel's Eigenfunction Matching Method (EMM) and Meylan's BEM/FEM approach (Section 3 above, and [?] as validation). This phase doesn't need new development from us – it's the machinery we lean on both to generate synthetic data (see the twin test) and, eventually, to interpret Phase 3's output.
2. **Single-sensor modal extraction.** Given only the real (or synthetic) single-point IMU time series, extract the system's dominant temporal dynamics – resonant frequencies and

damping – without knowing the underlying geometry, using some data-driven system identification method. I don’t want to commit to a specific method yet; this is genuinely an open choice I’d like to discuss. Candidates include:

- **Dynamic Mode Decomposition (DMD)** and its Hankel/delay-embedded variants, or the closely related **Koopman/HAVOK** framing, which I was initially leaning towards;
- **Output-only Stochastic Subspace Identification (SSI)** or the **Eigensystem Realization Algorithm (ERA)**, standard tools from the structural/mechanical system ID literature for extracting modal parameters from a driven system’s output alone;

One thing I have become more confident about is that Takens’ delay-embedding theorem, which underpins the DMD/HAVOK framing, is built for *autonomous* dynamical systems, whereas our floe is linearly driven by external, non-autonomous wave forcing. So if we do end up using a DMD-family method, that mismatch needs to be addressed explicitly (e.g. treating the incident wave as a known/measured input rather than pretending the system is self-contained), rather than applying Hankel-DMD off the shelf. This is really the main thing steering me away from a naive HAVOK approach and towards looking more seriously at SSI/ERA, but I haven’t committed either way.

3. **Eigenvalue-space parameter identification.** Whatever method Phase 2 settles on, its output is some set of *temporal* eigenvalues (poles) describing the identified linear system’s dynamics. Montiel/Meylan’s forward theory (Phase 1) is built around *spatial* eigenvalues – roots of the dispersion relation that depend on (R, D, E^v) . These aren’t the same object, so this phase is about using the existing dispersion relation to translate one into the other and back out (R, D, E^v) . This is the step I’m least sure how to handle cleanly, and where I most want input – to be clear, the goal is to adapt/apply the existing relationship correctly, not to derive new theory.

The synthetic twin test described below is specifically aimed at Phase 3: it lets us test whether the eigenvalue-space mapping, once derived, can actually recover known ground-truth parameters, before we trust it on real Phase 2 output from the cruise data.

1.7.1 Aside: What I Mean by a "Synthetic Twin" Test

I’ve referenced the idea of a synthetic twin test a few times now without really explaining it, so let me spell it out here since it’s going to be central to how I want to approach Phase 3.

The core problem is that we don’t have ground truth for the real IMU data. Once we run the inversion pipeline on the real buoy data and get out some estimate of (R, D, E^v) (or whatever reduced parameter set we settle on), we have no independent way of knowing whether those numbers are *correct*, only whether they’re *self-consistent* (e.g. the residual is small, the reconstructed spectrum looks plausible). That’s a weak check. It’s entirely possible for an inversion to converge to a stable, low-residual, wrong answer, especially given how underdetermined the single-sensor problem is (as discussed in the Meylan 2015 summary above).

A synthetic twin test sidesteps this by inverting on data where we already know the answer:

1. **Pick a ground-truth floe.** Choose a specific, known (R, D, E^v) , i.e. commit to a floe radius/geometry and a viscoelastic material model.

2. **Generate synthetic data with the forward model.** Feed a realistic incident wave spectrum through Montiel’s EMM (or the modal system in Eq. 1.33 above, depending on geometry), and evaluate the resulting displacement field at a single point corresponding to where the IMU would sit on the floe. This gives a synthetic single-sensor time series, built entirely from known physics with a known answer baked in.
3. **Add realistic noise/imperfections.** Corrupt the synthetic signal with something resembling the real buoy’s sensor noise, sampling characteristics, and (importantly) a realistic incident wave spectrum rather than a single clean tone, since the real ocean forcing is broadband, not monochromatic.
4. **Run the full inversion pipeline on this synthetic signal,** exactly as we intend to run it on the real data: SSI/ERA/DMD/POD or whatever other data-driven method for modal extraction, then the eigenvalue-space mapping back to (R, D, E^v) .
5. **Compare recovered parameters to the ground truth we chose in step 1.**

If the recovered parameters don’t match the ground truth even in this fully controlled setting (no real-world confounds, no model mismatch, we generated the data with the exact same model we’re inverting with), that tells us the identifiability problem is fundamental to the single-sensor setup itself, not an artefact of real-data messiness. This is a much cheaper and more diagnostic way to find that out than discovering it after weeks of fighting with the real cruise data.

It also lets us probe identifiability systematically in a way we can’t with real data: sweep across different ground-truth (R, D, E^v) combinations and see which regions of parameter space are well-recovered versus degenerate (e.g. maybe D and E^v trade off against each other in a way that’s impossible to disentangle from a single sensor, in which case we’d know to report a joint constraint rather than independent estimates). We could also use it to answer the question I raised at the end of the Meylan 2015 summary – sweep floe size/wave conditions and see empirically how many modes are actually recoverable from one sensor, rather than guessing.

1.8 All of the above is a proposal not a final product!

I am not 100% sure of the above pipeline. It is something that needs proper brainstorming! The last 2 weeks have been spent mainly on literature reviews. I am veeeeerry open to discuss proper ideas when we meet!